

Lab 1

Handling and Injection of Experimental Animals

**MMUNOLOGY & SEROLOGY LABORATORY
(BT336)**

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Humoral and Cellular Immunity

Adaptive immune responses have historically been divided into two separate arms of defense:

1. B-cell-mediated or humoral immune responses

- B cells are initially activated to secrete antibodies after the binding of antigens to antigen-specific membrane immunoglobulin (Ig) molecules (B-cell receptors [BCRs]), which are expressed by these cells.

2. T-cell-mediated or cellular responses

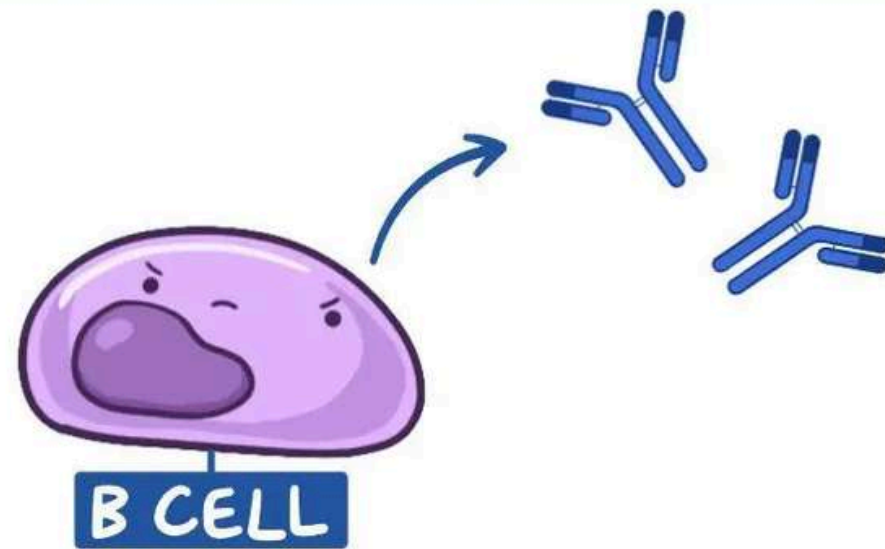
- The elimination of a foreign antigen such as a pathogenic microbe can be an activated T cell expressing pathogen-specific T-cell receptors (TCR) or it can be a phagocytic cell that gets activated by innate receptors that they express, and the cytokines produced by activated T cells

Humoral and Cellular Immunity

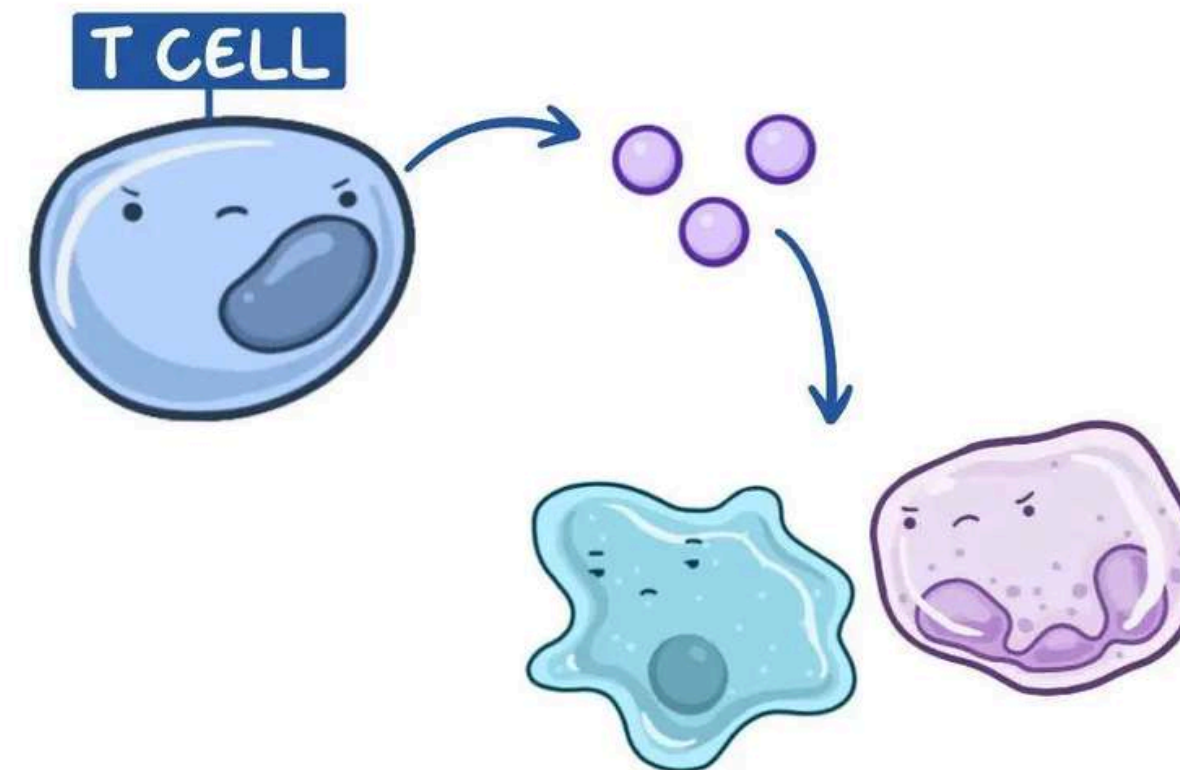
ACTIVE IMMUNE SYSTEM

- * SPECIFIC & SLOWER RESPONSE
- * T & B LYMPHOCYTES

HUMORAL RESPONSE

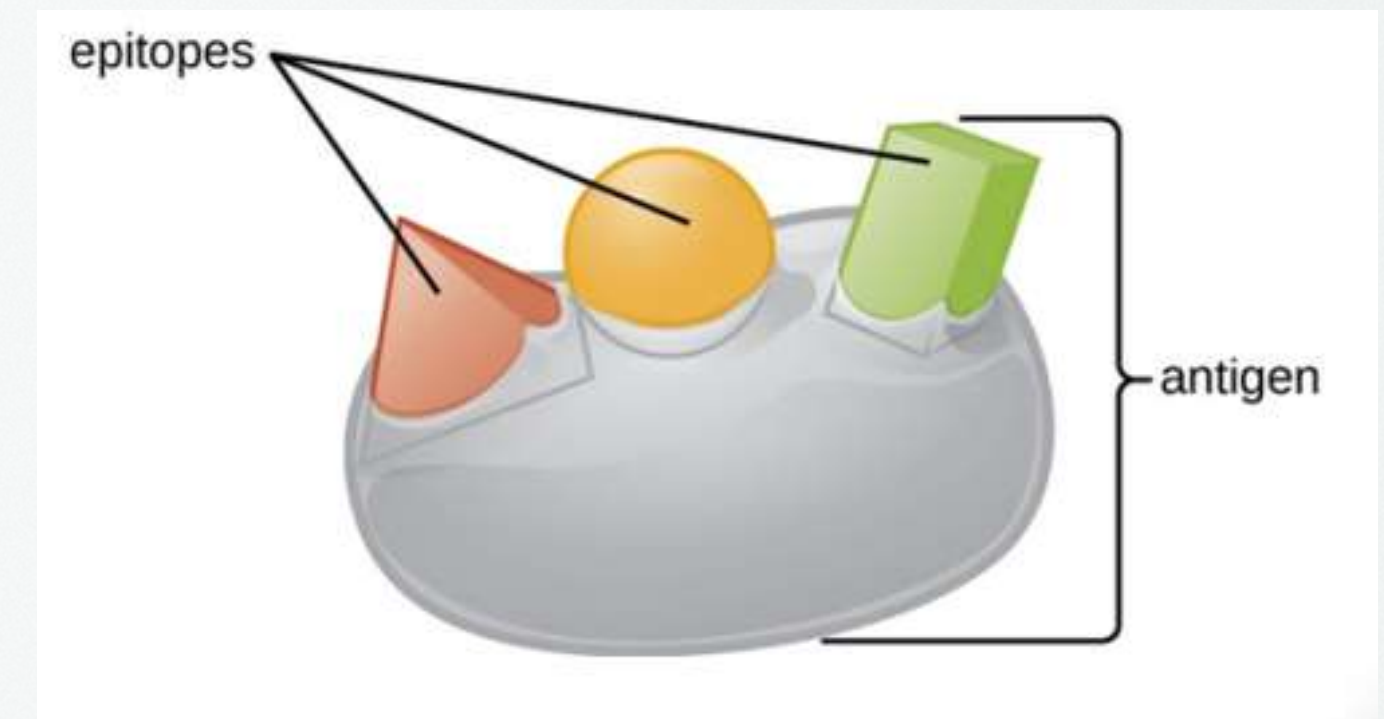


CELL-MEDIATED RESPONSE



Immunological Keys

- **Antigen:** any foreign substances that could or could not stimulate the immune system.
- **Immunogen:** any foreign substance that produces a humoral or cell-mediated immune response.
 - All immunogens are antigens, but all antigens may not be immunogens
- **Epitope (antigenic determinant):** the part of an Immunogen that is recognized by an antibody, B-cell receptor, or T-cell receptor



Immunization

- Immunization is the process by which an individual is made immune or resistant to an infectious disease, typically through the administration of a vaccine.



Types of Immunization

1. Active Immunization – The body produces its own immune response after exposure to an antigen.

- **Natural:** Occurs when a person is exposed to a pathogen and develops immunity (e.g., recovering from an infection).
- **Artificial:** Achieved through vaccines, which contain weakened, killed, or parts of pathogens to stimulate an immune response.

2. Passive Immunization – Immunity is provided by transferring antibodies from another individual.

- **Natural:** Occurs when maternal antibodies are passed to a baby through the placenta or breast milk.
- **Artificial:** Involves injecting pre-formed antibodies (e.g., monoclonal antibodies, antitoxins).

Animal Immunization

Animal immunization—it involves introducing an immunogen (such as a vaccine or antigen) into an animal or human to stimulate a specific humoral (antibody-mediated) and/or cellular (T-cell-mediated) immune response.

Immunogen could be in two forms:

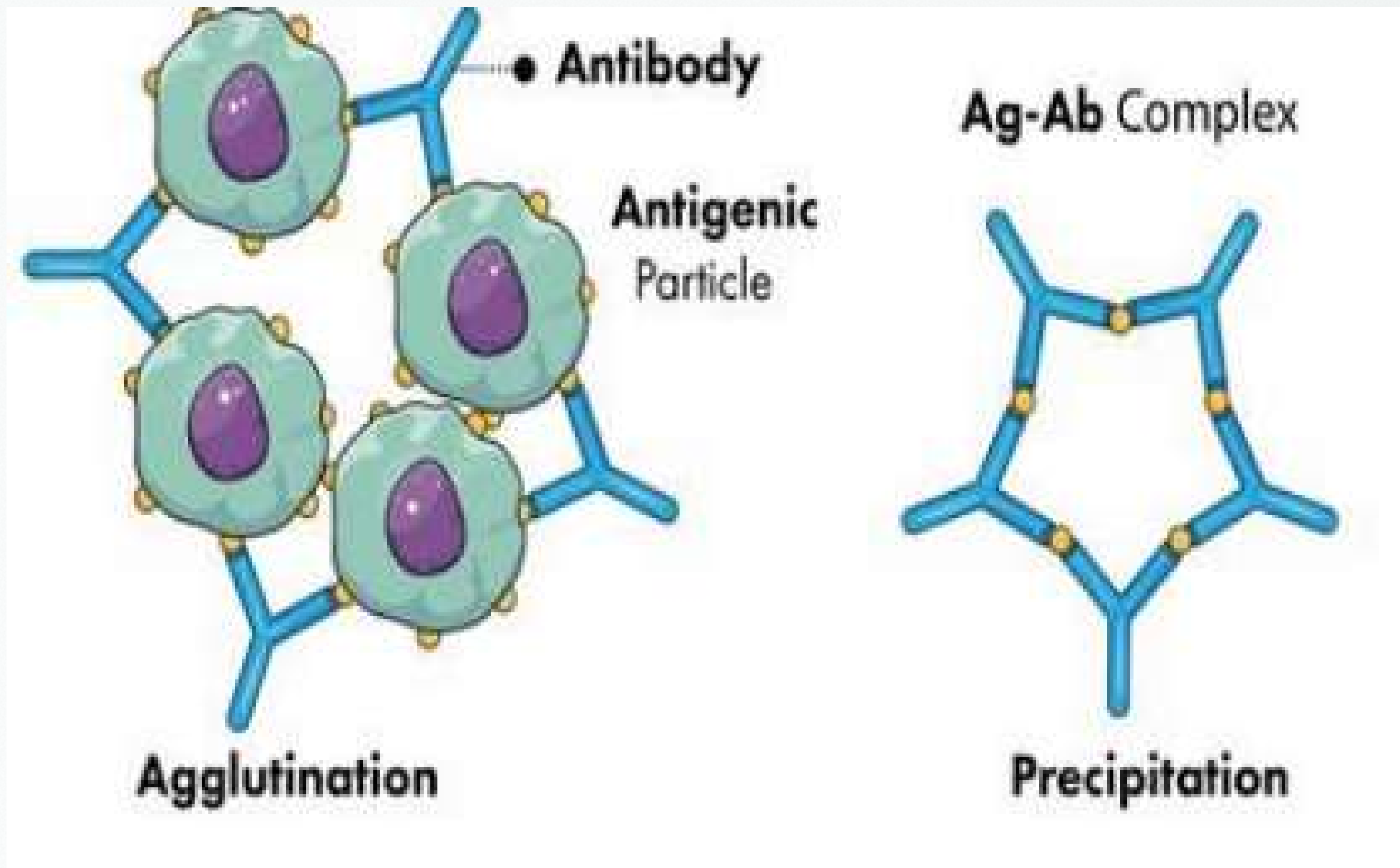
1. **Soluble** e.g., soluble protein (Ova albumin), Bovine serum albumin "BSA" and Bovine gamma globulin "BGG")

Soluble Ag + Soluble Ab $\longrightarrow$ Precipitation reaction

2. **Particles** e.g., whole bacterial cells killed by heat or formalin or erythrocyte suspension.

Particulate Ag + Soluble Ab $\longrightarrow$ Agglutination reaction

Animal Immunization



Immunogens Characteristics

1. Chemical Composition

- The biochemical nature of the immunogen significantly affects its immunogenicity.
- **Proteins:** The most potent immunogens due to their structural complexity and diverse epitopes.
- **Polysaccharides:** Can be immunogenic, especially if they are complex and part of a pathogen's surface (e.g., bacterial capsules).
- **Lipids and Nucleic Acids:** Poorly immunogenic on their own but can induce an immune response when conjugated to proteins

2. Molecular Size

- Generally, **larger molecules** (>10 kDa) are more immunogenic than smaller ones.
- **Small molecules** (haptens, <1 kDa) are not immunogenic on their own but can induce an immune response when attached to a larger carrier molecule.

Immunogens Characteristics

3. Chemical Complexity

- The more complex a molecule is, the greater the number of antigenic determinants (epitopes) it possesses.
- Complex molecules like proteins with multiple secondary and tertiary structures stimulate both B cells and T cells, leading to a stronger response.
- Homopolymers (e.g., synthetic polymers with repeating identical units) tend to be poorly immunogenic.

4. Purity and Sterility

- The immunogen must be free from microbial contamination to prevent unwanted immune responses or infections.

5. Free of Toxicity

- The immunogen should not contain toxic components that could cause harm.

Immunogens Characteristics

6. Suitable pH

- The immunogen should be stable within the physiological pH range (pH 7.2–7.4) to maintain its structural integrity and biological function.
- Extreme pH levels can denature proteins and alter epitopes, reducing immunogenicity.

7. Appropriate Dose

- The immune response is dose-dependent:
- Too low a dose may lead to tolerance or insufficient immune activation.
- Too high a dose can lead to immune suppression or excessive inflammation.

Routes of Immunogen Administration

- **The route of administration significantly affects the immune response induced by an immunogen.**
- **The route of administration largely depends on the property of the test substance and the objective of the experiment.**
- **All administration should be performed with knowledge of the chemical and physical characteristics of the substance.**
- **Different routes influence antigen uptake, processing, and the type of immune response generated (humoral vs. cellular immunity).**

Routes of Immunogen Administration

1. Intramuscular (IM) Injection

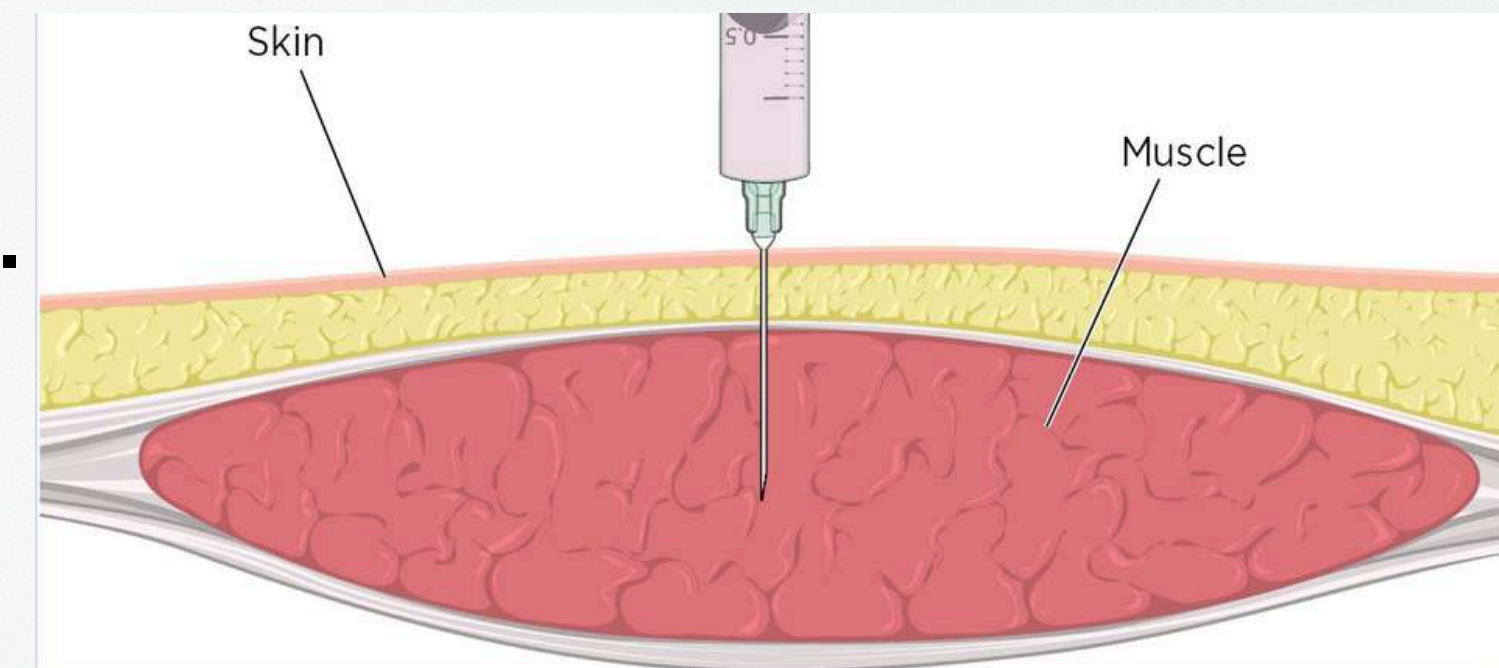
- Intramuscular (IM) administration involves injecting therapeutic agents directly into the muscle tissue.

Advantages

- Strong immune response, including both humoral (antibody-mediated) and cellular (T-cell-mediated) immunity.
- Slower antigen clearance, leading to longer-lasting immunity.

Disadvantages

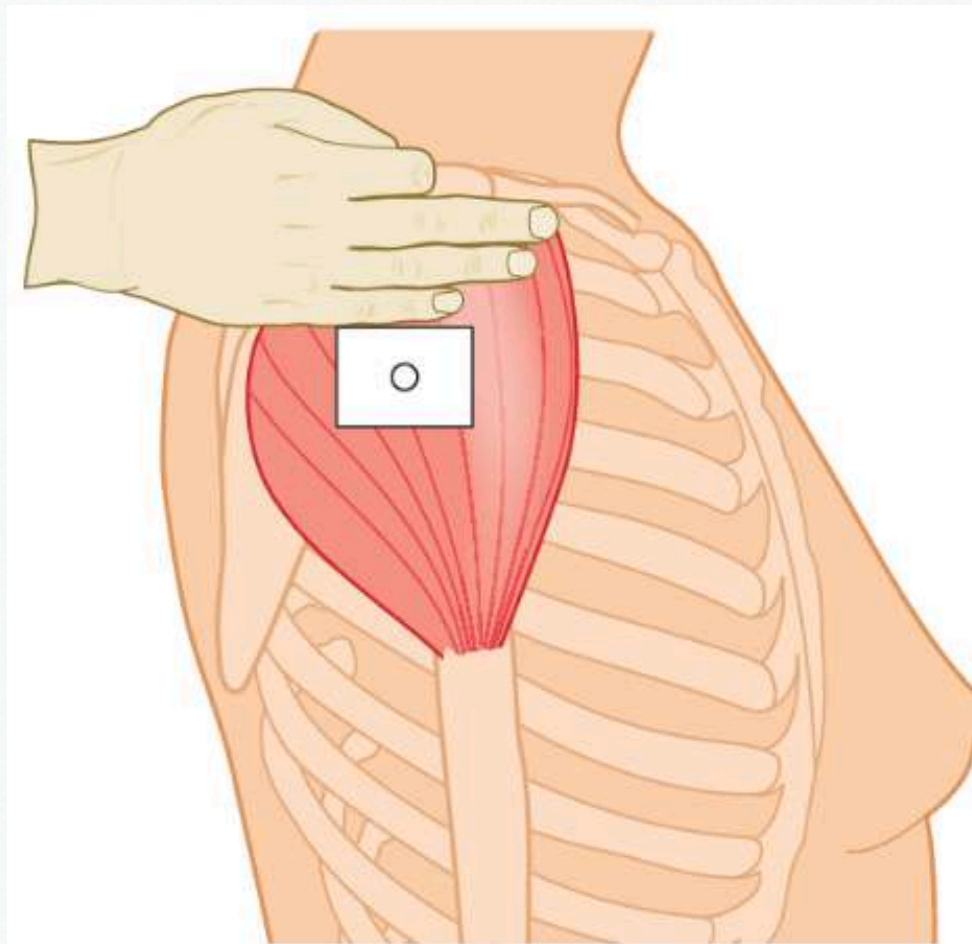
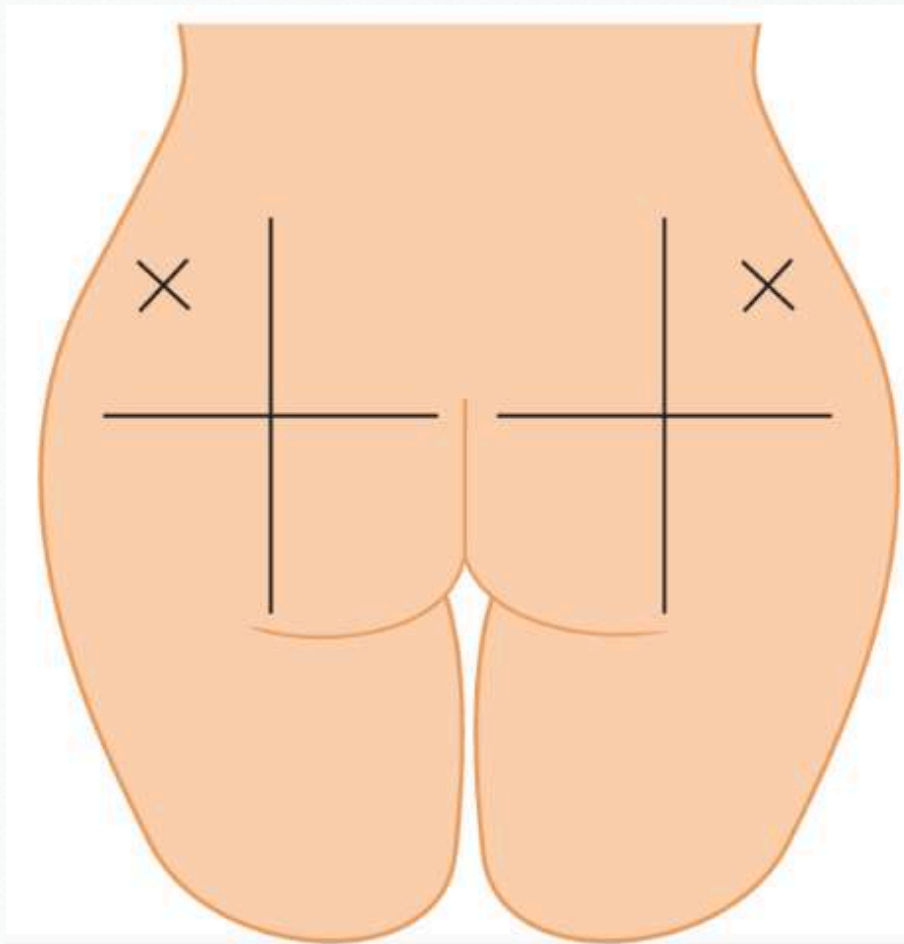
- Can cause local pain, swelling, and muscle soreness.



The needle should be inserted at 90 degrees and penetrate the muscle layer

Routes of Immunogen Administration

1. Intramuscular (IM) Injection



Routes of Immunogen Administration

2. Subcutaneous (SC) Injection

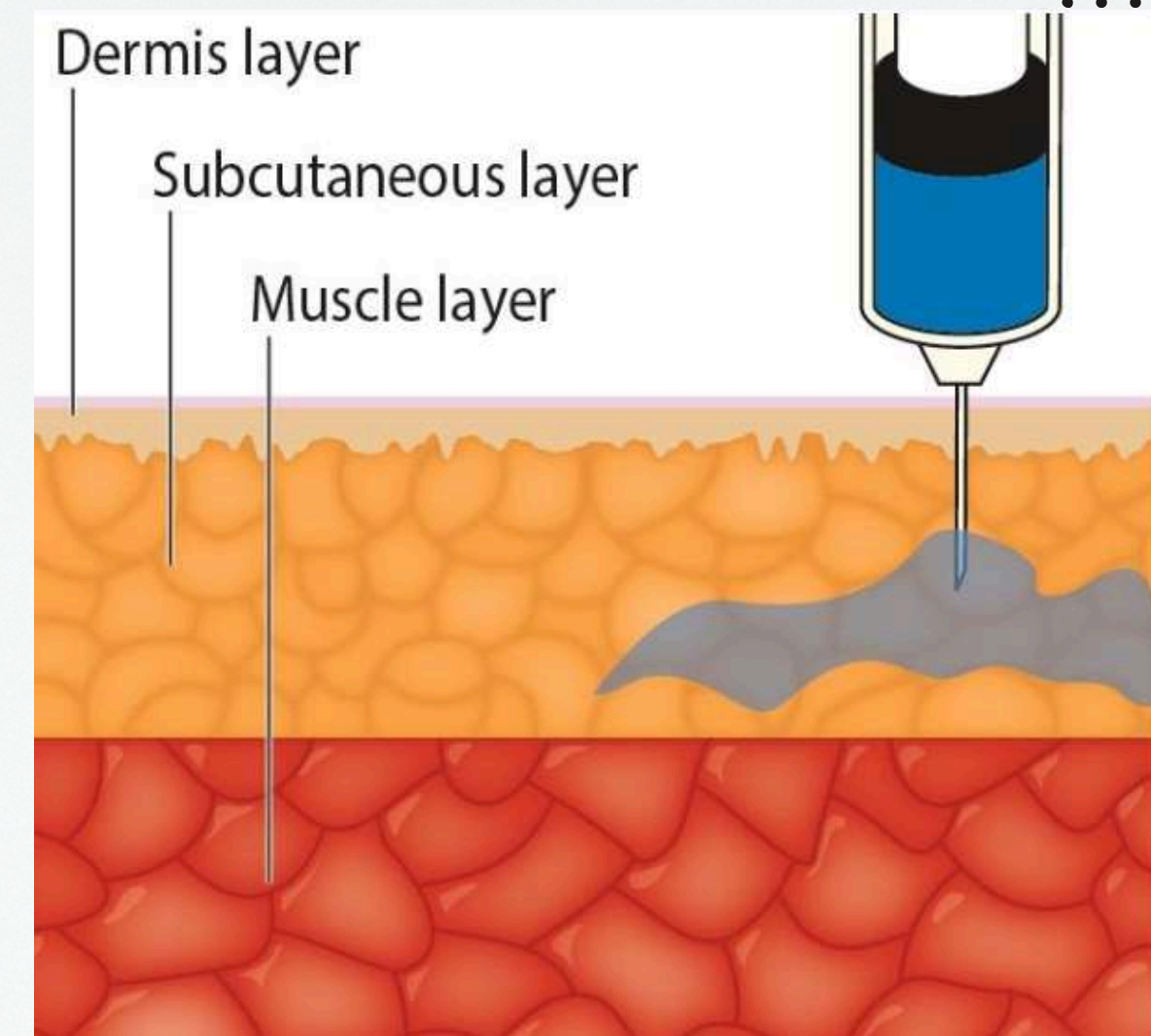
- Subcutaneous (SC) administration is a method of delivering substances by injecting them into the subcutaneous tissue, which is the layer of tissue just beneath the skin.

Advantages

- Slower absorption than IM, leading to prolonged antigen exposure.

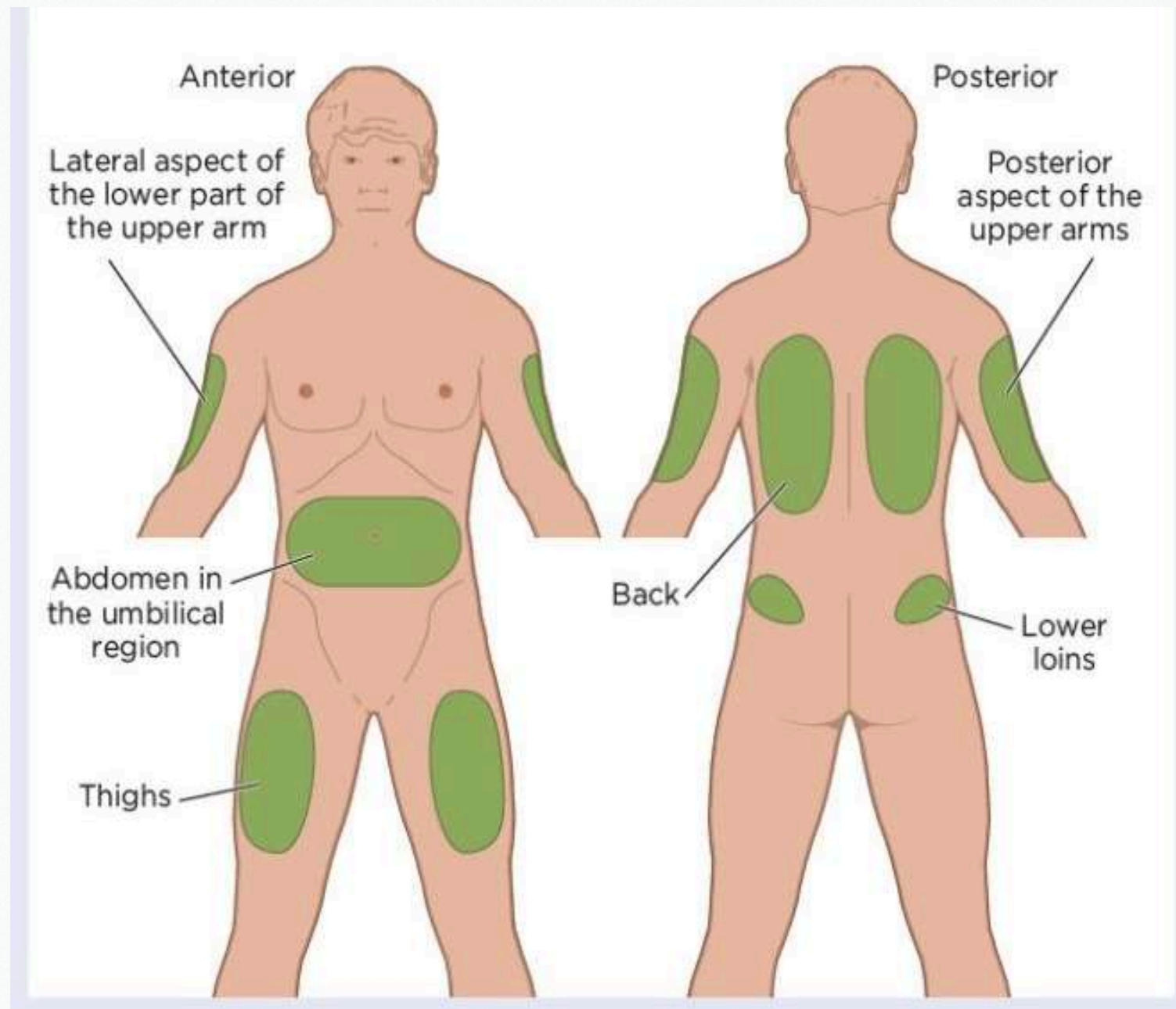
Disadvantages

- May cause local irritation and swelling.
- Slightly less efficient than IM due to less vascularization.



Routes of Immunogen Administration

2. Subcutaneous (SC) Injection



Routes of Immunogen Administration

3. Intradermal (ID) Injection

- An intradermal (ID) injection is a method of delivering a small volume of liquid into the dermis, the layer of skin just beneath the epidermis.

Advantages

- The high density of dendritic cells (APCs) in the skin, leads to a strong immune response.
- Requires a lower dose compared to IM or SC, making it more cost-effective.

Disadvantages

- More difficult to administer correctly; improper technique can lead to failed immunization.
- Can cause local irritation and skin reactions.

Routes of Immunogen Administration

3. Intradermal (ID) Injection



Routes of Immunogen Administration

4. Oral (PO) Administration

- Oral administration is a common method of delivering medications, supplements, or other substances into the body through the mouth.

Advantages

- Non-invasive and easy to administer, making it ideal for mass vaccination programs.

Disadvantages

- Antigens may be degraded by stomach acid and digestive enzymes, requiring special formulations

Routes of Immunogen Administration

3. Oral (ID) Injection



Routes of Immunogen Administration

5. Intranasal (IN) Administration

- Intranasal (IN) administration is a non-invasive method of delivering medications through the nasal mucosa for systemic absorption or local effects.

Advantages

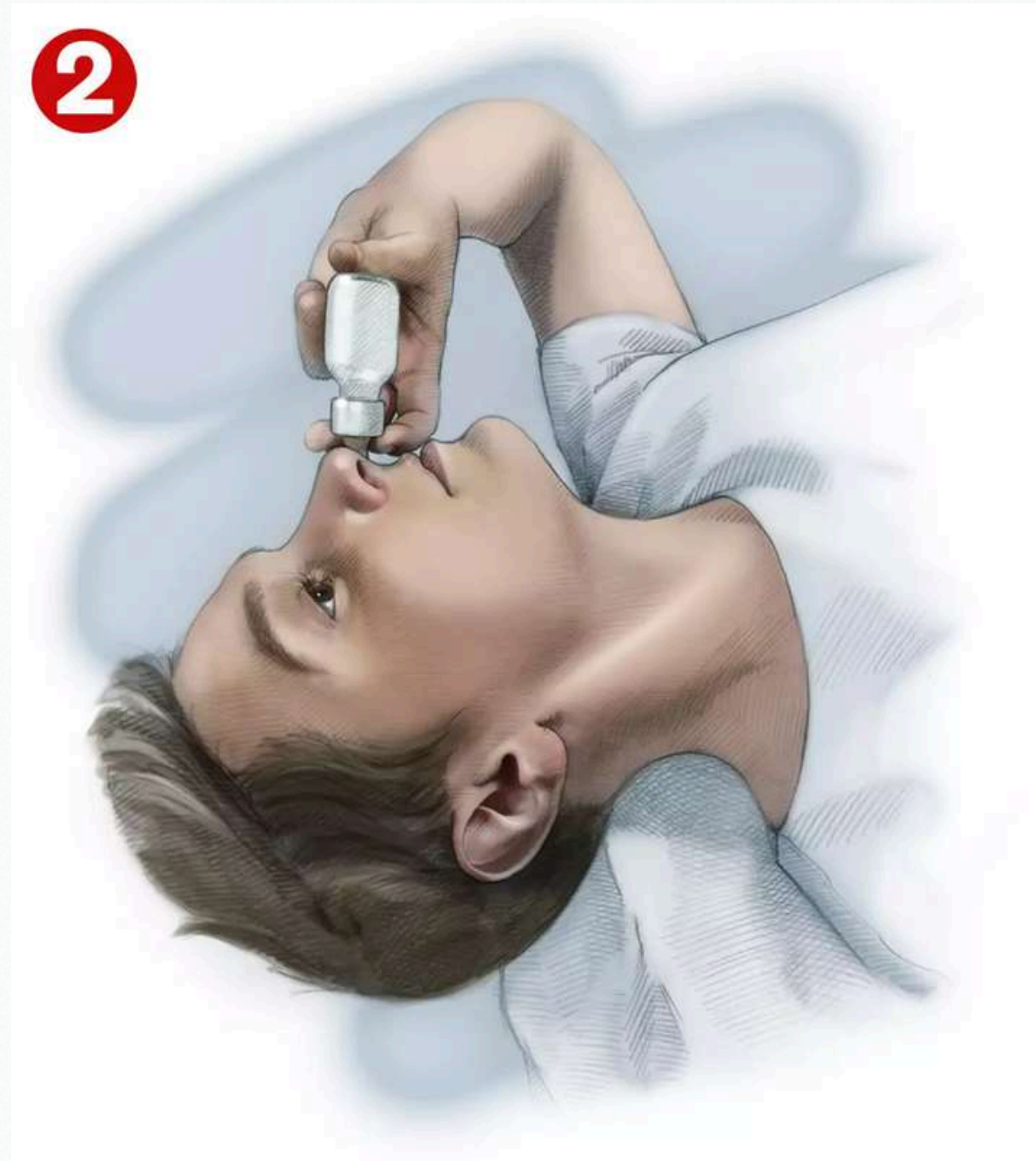
- Induces mucosal immunity in the respiratory tract, protecting against airborne pathogens.
- Needle-free, making it more acceptable, especially for children.

Disadvantages

- Can cause nasal irritation and mild side effects like congestion.
- Less consistent immune response compared to injectable vaccines.

Routes of Immunogen Administration

5. Intranasal (IN) Administration



Routes of Immunogen Administration

6. Intravenous (IV) Administration

- Intravenous (IV) administration involves delivering a drug, fluid, or vaccine directly into the bloodstream through a vein. This method is considered the most efficient and rapid route of administration

Advantages

- **100% Bioavailability** – Unlike oral or intramuscular routes, IV administration ensures that all of the drug reaches circulation without loss due to metabolism or poor absorption.
- IV administration allows for exact dosing with minimal variability in drug absorption.
- Since the drug is delivered directly into the bloodstream, it bypasses absorption barriers, leading to immediate effects.

Routes of Immunogen Administration

6. Intravenous (IV) Administration

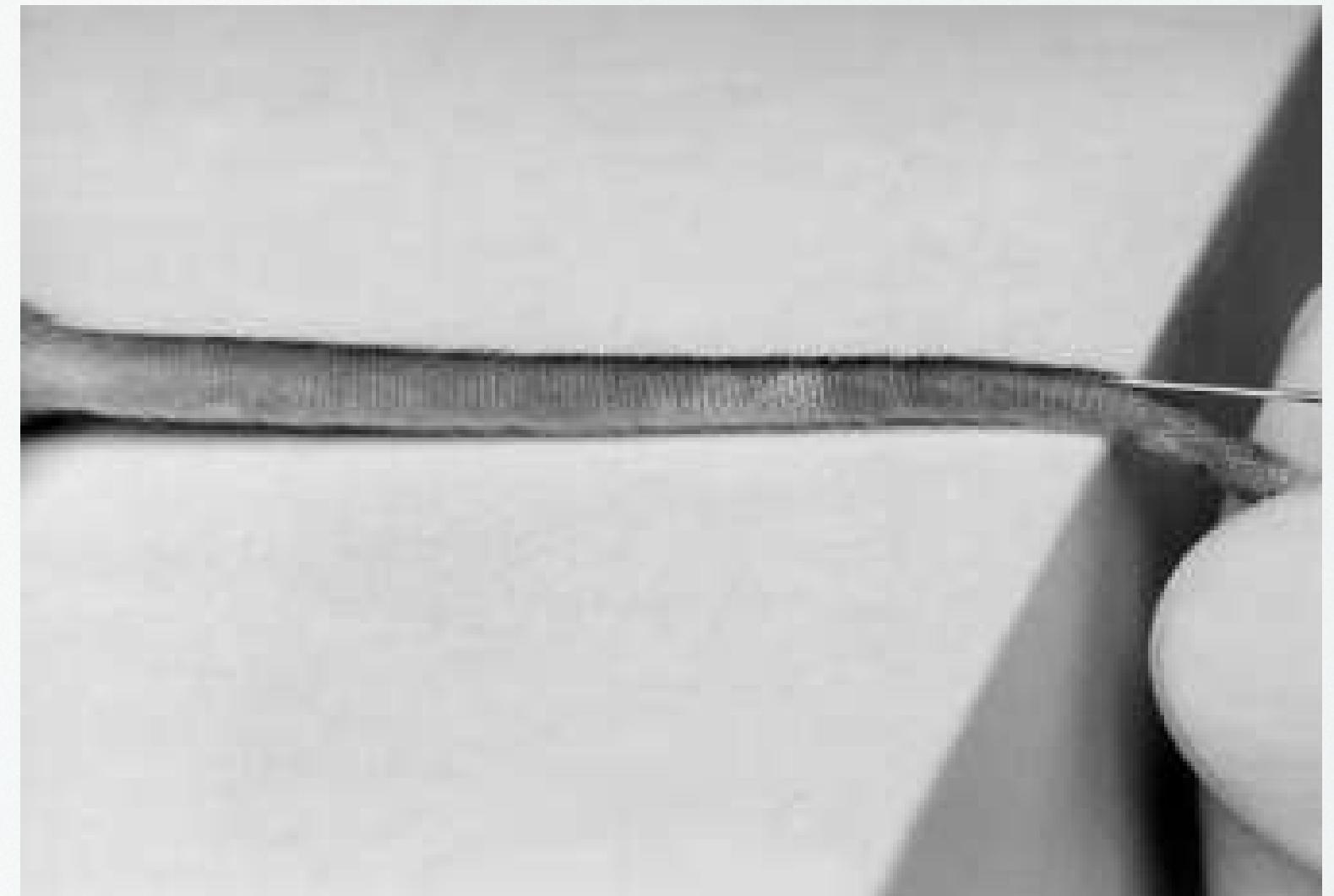
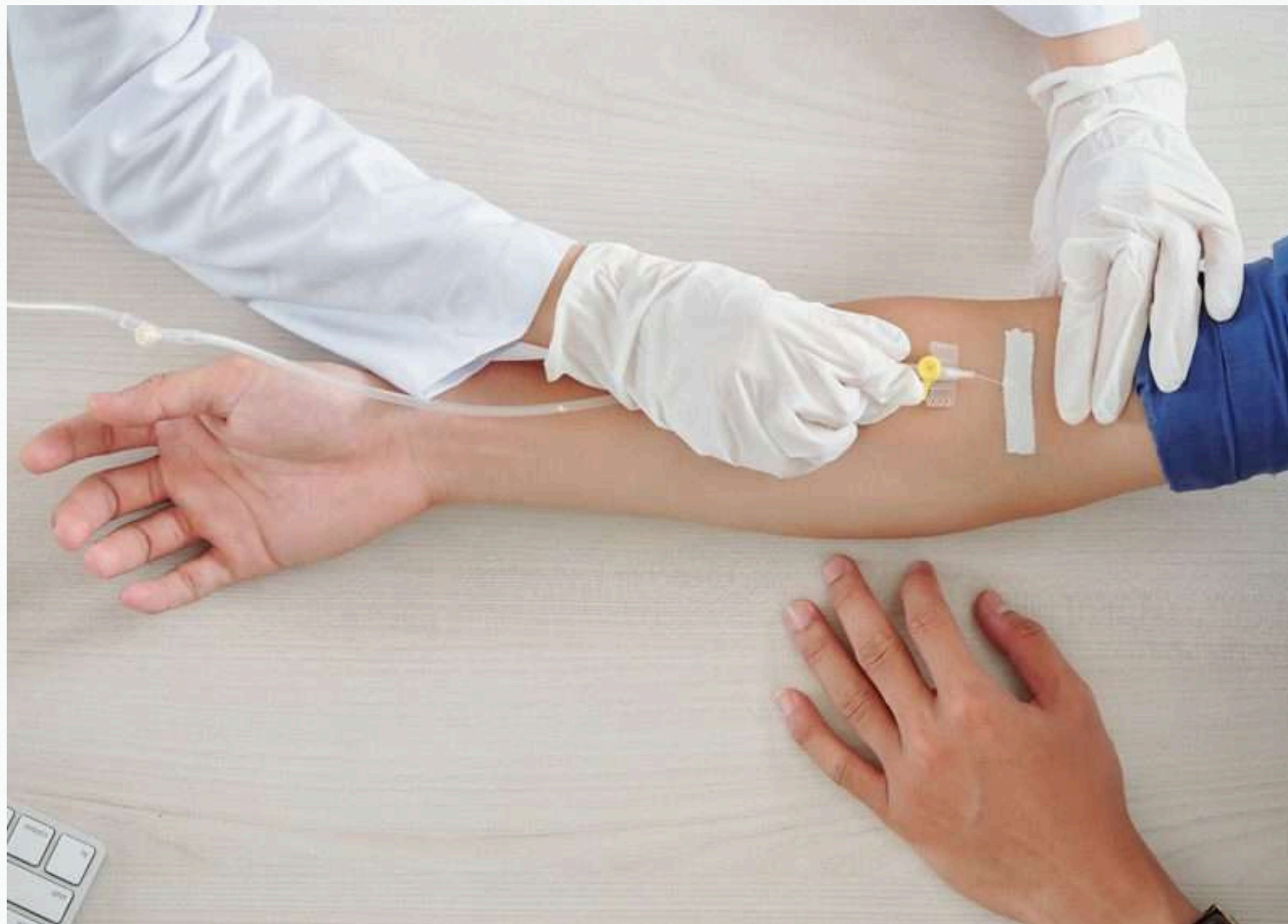
- Intravenous (IV) administration involves delivering a drug, fluid, or vaccine directly into the bloodstream through a vein. This method is considered the most efficient and rapid route of administration

Disadvantages

- Invasive – Requires venous access, which may be difficult in some patients.
- Risk of Infection – Improper handling can introduce infections into the bloodstream.
- Potential for Adverse Reactions – Since the drug enters the bloodstream immediately, side effects can appear rapidly and intensely.

Routes of Immunogen Administration

6. Intravenous (IV) Administration



Routes of Immunogen Administration

7. Rectal (PR) Administration

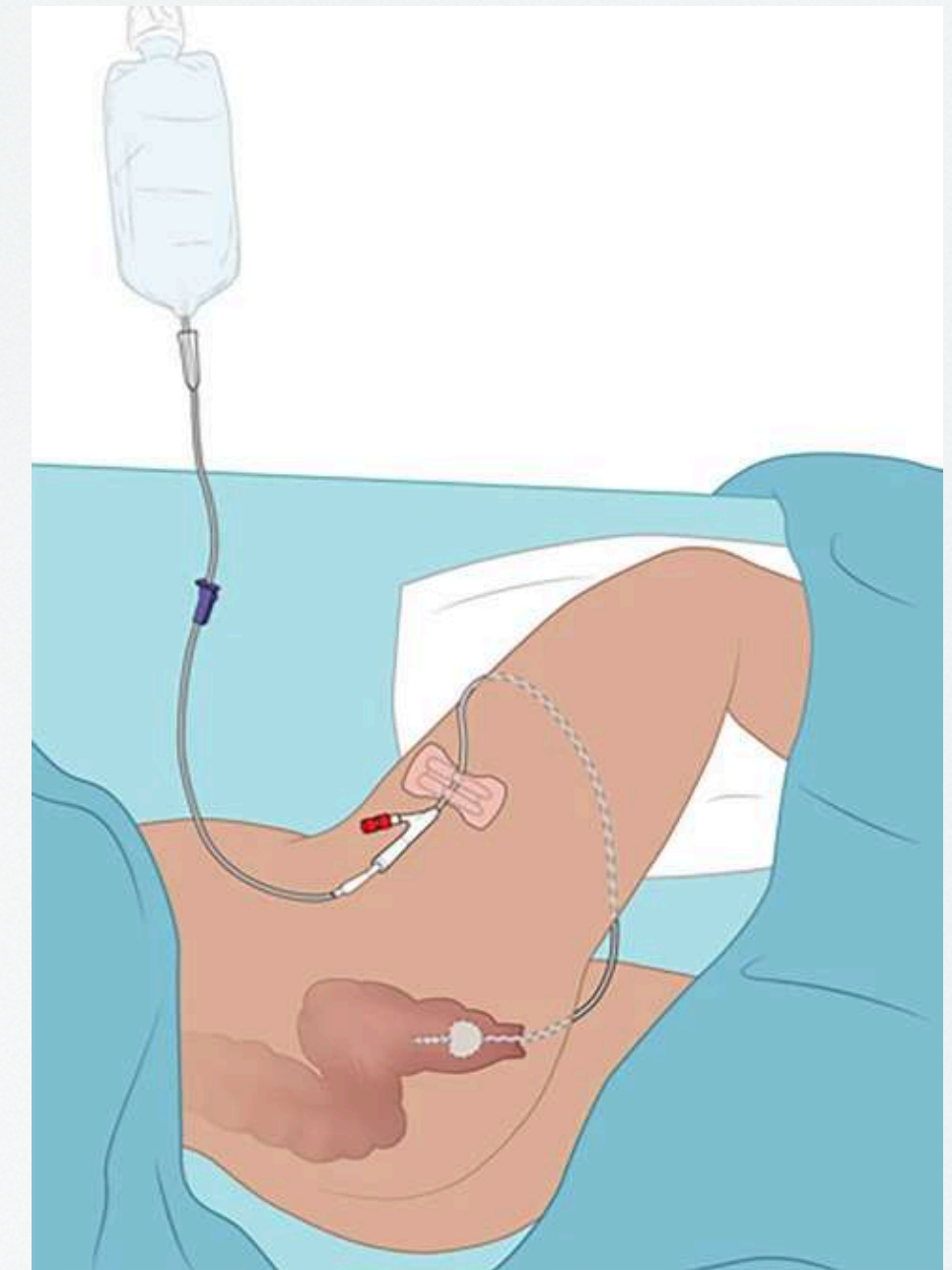
- Rectal (PR) administration involves delivering medications through the rectum

Advantages

- Induces mucosal and systemic immunity.

Disadvantages

- Less commonly used, and patient compliance is low.



Routes of Immunogen Administration

8. Intraperitoneal (IP) Administration

- Intraperitoneal (IP) administration involves injecting medications or fluids directly into the peritoneal cavity, the space within the abdomen that houses organs such as the intestines, liver, and stomach

Advantages

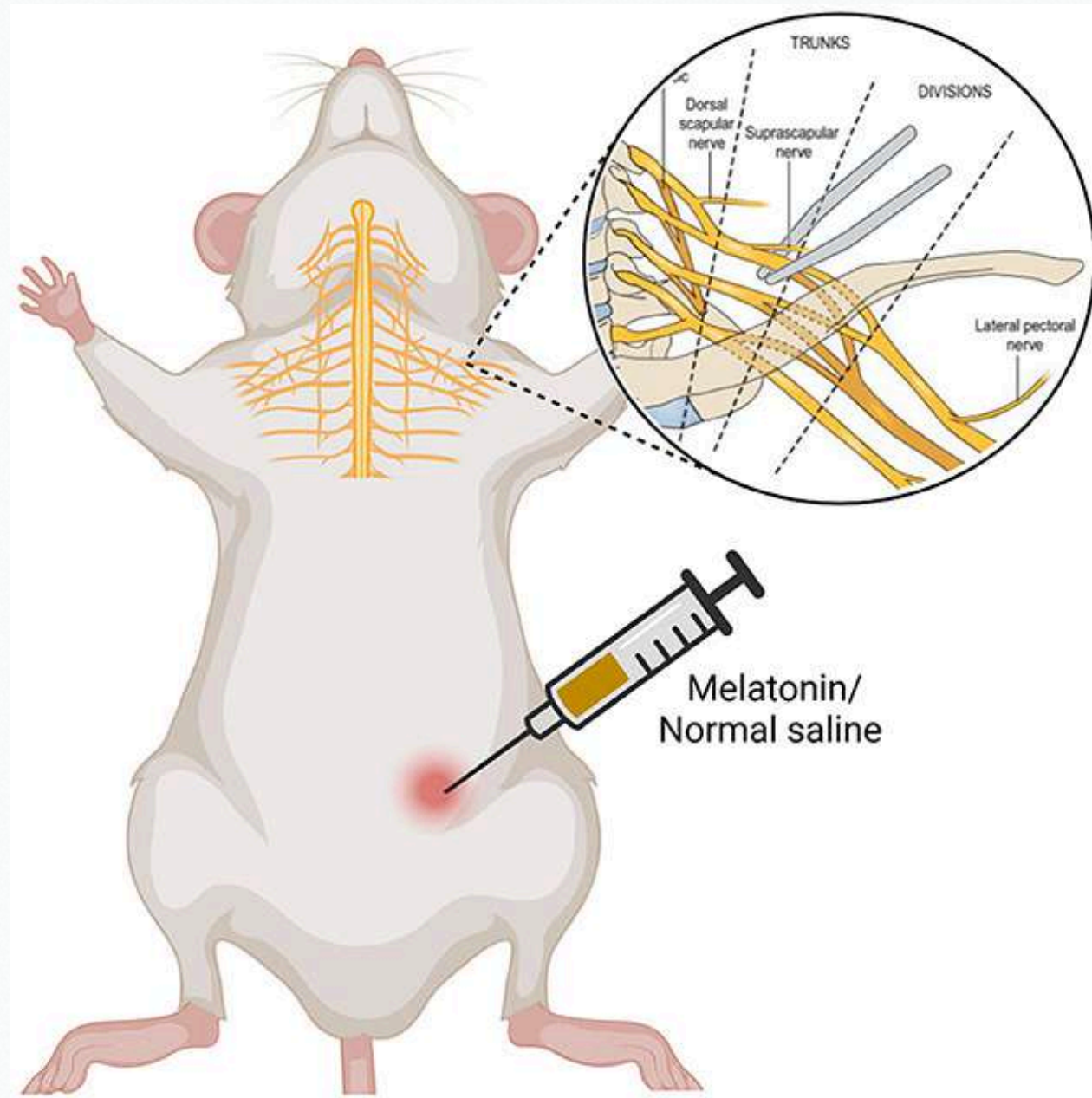
- Faster absorption than subcutaneous or intramuscular injections.
- Stronger immune response compared to oral or nasal routes.
- Can be used for adjuvant administration in experimental models.

Disadvantages

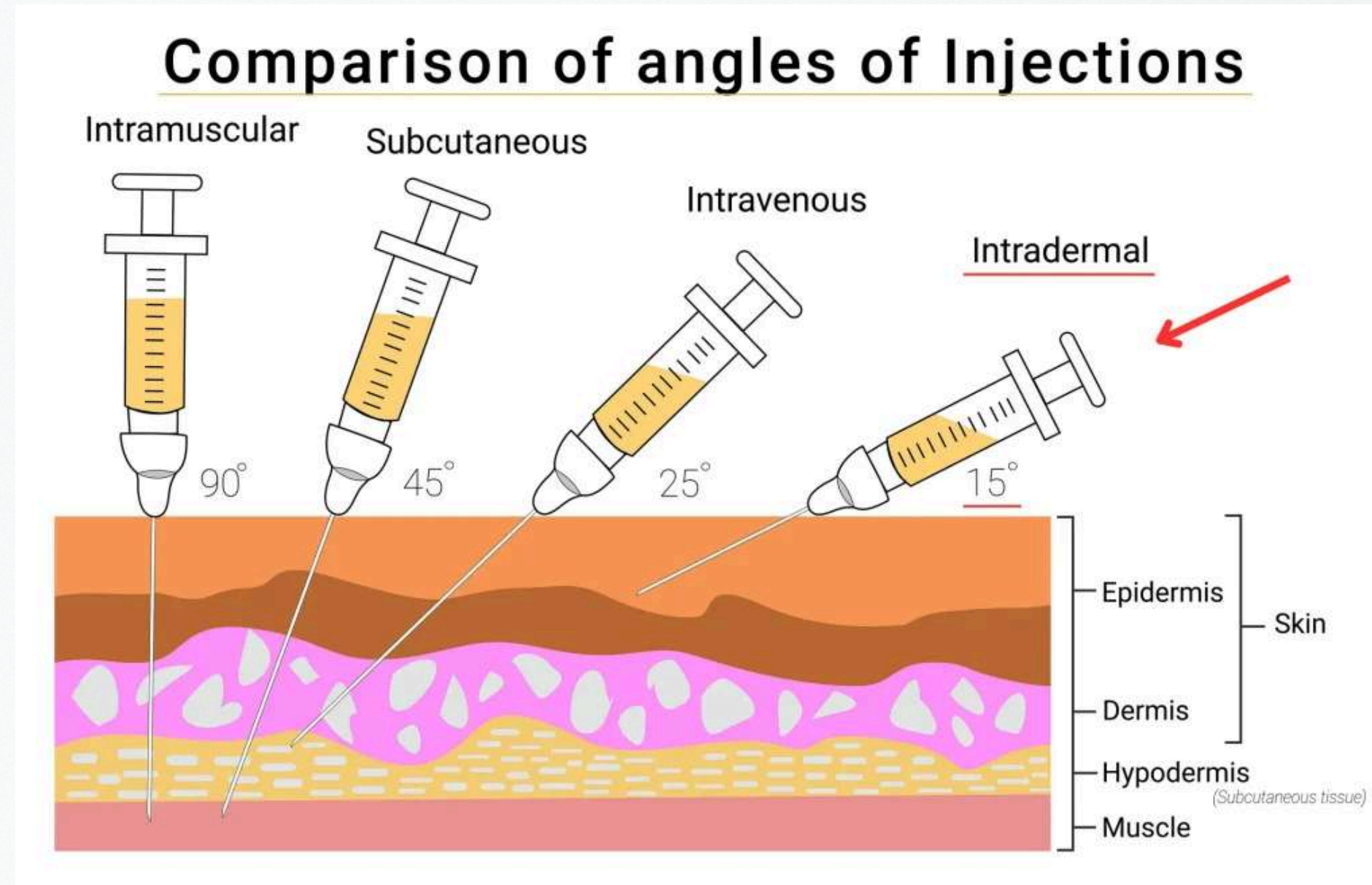
- Not commonly used in humans due to potential risks, including infection and organ damage.
- Difficult to administer safely compared to IM or SC injections.
- Higher chance of peritonitis (inflammation of the peritoneal lining)

Routes of Immunogen Administration

8. Intraperitoneal (IP) Administration



Routes of Immunogen Administration



Adjuvant

- **Adjuvants are substances that enhance the immune response to an antigen when included in a vaccine or immunotherapy.**
- **They work by boosting, prolonging, or modulating the immune system's reaction, making vaccines more effective.**
- **The adjuvant is used if only a small amount of soluble immunogen is available or if the material is weakly immunogenic.**
- **The adjuvant-immunogen mixture is given IM, IP, SC, and ID but not IV,**
- **Only native soluble immunogen can be given IV**

Mechanism of Action of Adjuvants

Adjuvants work by stimulating the innate immune system to enhance the adaptive immune response to an antigen. They achieve this through several mechanisms:

1. Activation of Innate Immunity

2. Depot Formation (Antigen Retention & Slow Release)

- Some adjuvants create a local depot (reservoir) at the injection site, prolonging antigen exposure to immune cells.
- This enhances immune cell recruitment and antigen uptake by antigen-presenting cells (APCs).

3. Enhanced Antigen Uptake & Presentation by APCs: Adjuvants increase antigen uptake by dendritic cells (DCs) and macrophages, improving MHC class I and II presentation to T cells.

4. Cytokine & Chemokine Production (Inflammatory Response)

Types of Adjuvants

1 Aluminum-based adjuvants (Alum):

- Most commonly used in human vaccines.
- Induces strong antibody responses.

2 Saponin-based adjuvants

- Derived from plant extracts (e.g., saponaria tree).
- Induce strong Th1 and Th2 immune responses.

3 Microbial substances

- e.g., Bacterial cell wall

4 Oil-in-water emulsions

- These mixtures of oil and water have proven their effectiveness in generating strong immune responses. Stimulate both humoral and cellular immunity.
 - **Complete Freund's adjuvant (C.E.A):** Freund's adjuvant with Mycobacterium Antigens (M. tuberculosis. or M. bytyricum).
 - **Incomplete Freund's adjuvant (I.F.A):** Freund's adjuvant without Mycobacterium antigens.

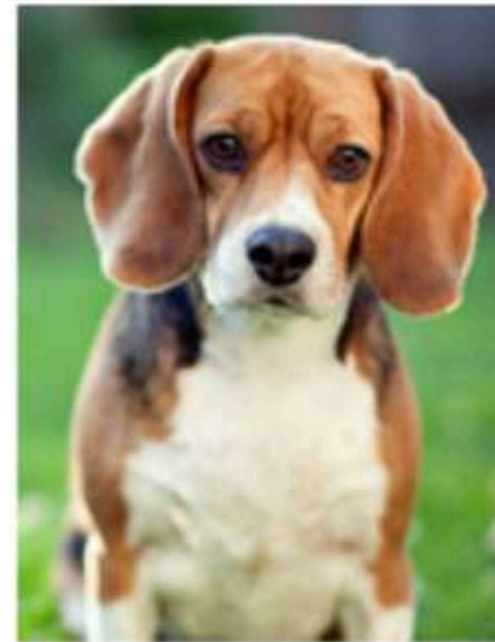
Experimental Animal Models & Their Uses



Baboon



Cats



Dogs



Mouse



Rat



Guinea pigs



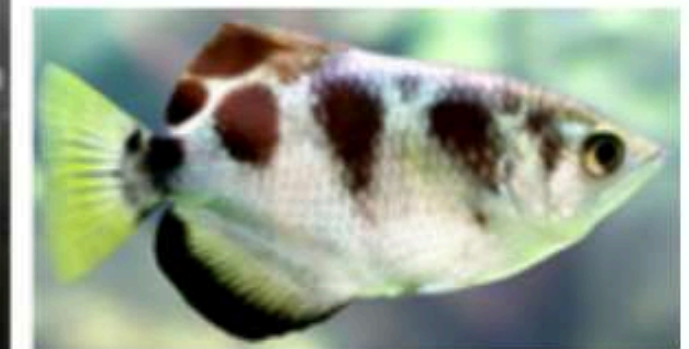
Hamster



Rabbit



Chicken



Fish

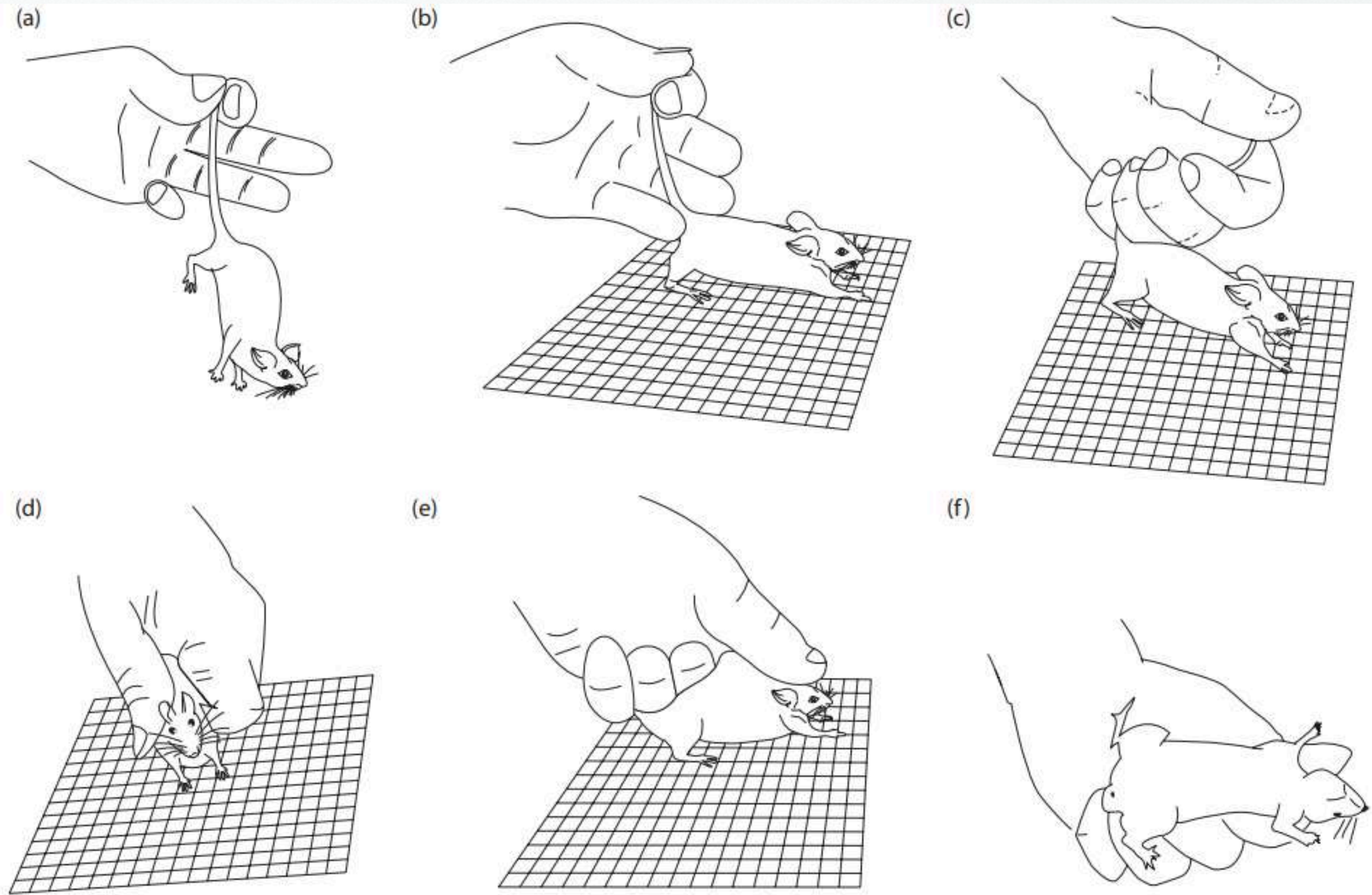
Experimental Animal Models & Their Uses

Animal	Scientific Name	Genetic Similarity to Humans (%)	Research Applications
Chimpanzees	<i>Pan troglodytes</i>	98–99%	Infectious diseases (HIV, Hepatitis), Neuroscience
Rhesus Macaque	<i>Macaca mulatta</i>	93–95%	Vaccine development, Immunology, Neurology
Mice	<i>Mus musculus</i>	85–90%	Genetics, Cancer, Drug testing
Rats	<i>Rattus norvegicus</i>	80–85%	Toxicology, Neurology, Cardiovascular research
Guinea Pigs	<i>Cavia porcellus</i>	70–75%	Immunology, Respiratory diseases
Dogs	<i>Canis lupus familiaris</i>	82–84%	Cardiovascular studies, Pharmacology
Pigs	<i>Sus scrofa</i>	85–90%	Organ transplantation, Metabolic diseases
Zebrafish	<i>Danio rerio</i>	70–75%	Developmental biology, Genetics
Rabbits	<i>Oryctolagus cuniculus</i>	80–85%	Antibody production, Vaccine testing, Eye irritation tests

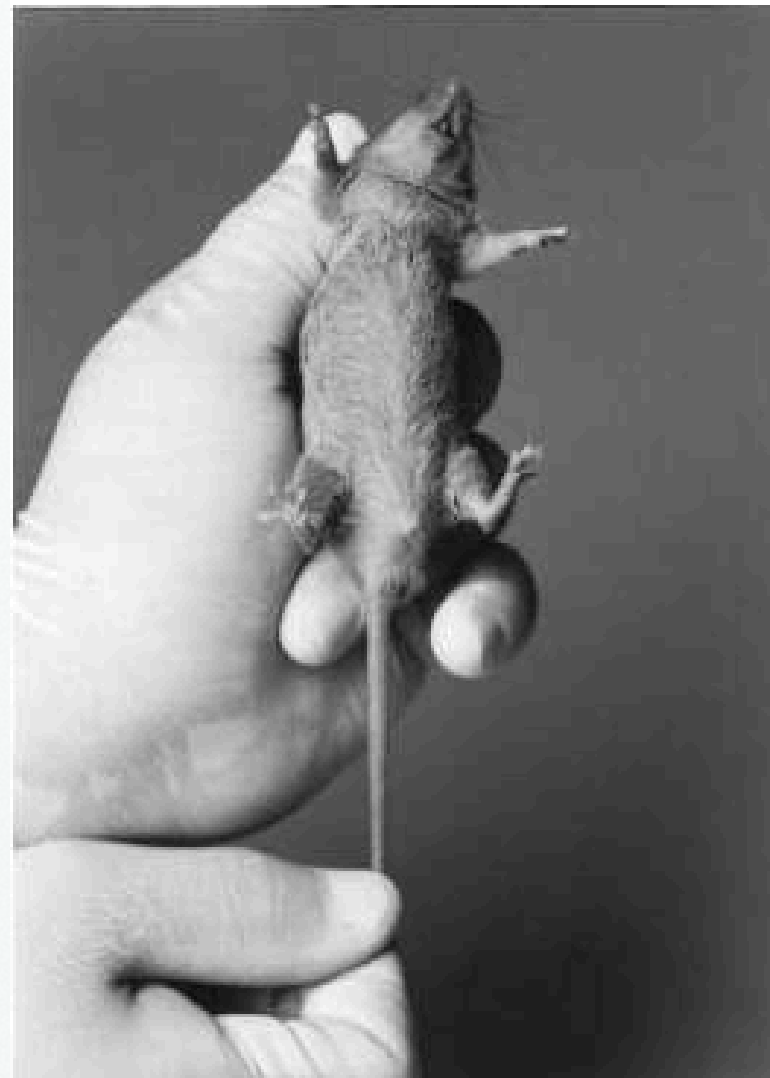
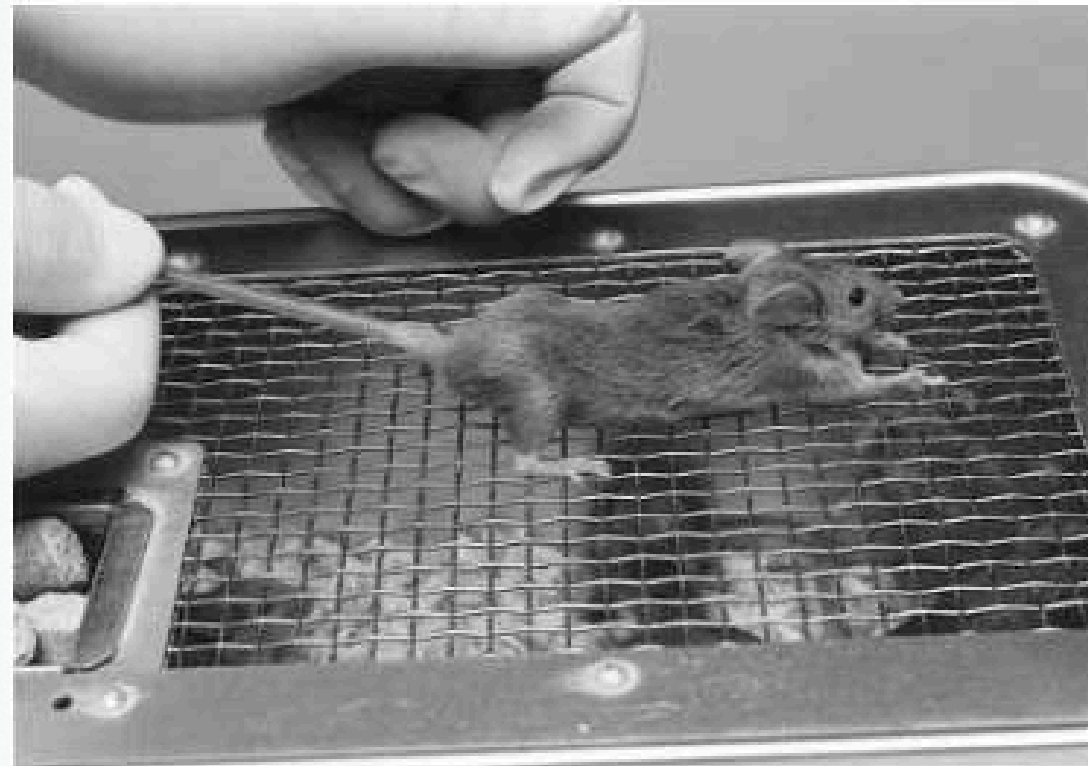
Handling of mice

- **Hold one mouse by the tip of the tail and let it walk on the bench; the mouse will not bite unless it is excited.**
- **The instructor will demonstrate gripping the mouse and holding it using the thumb, index, and middle fingers on his left hand.**
- **Do that yourself and practice it.**

Handling of mice



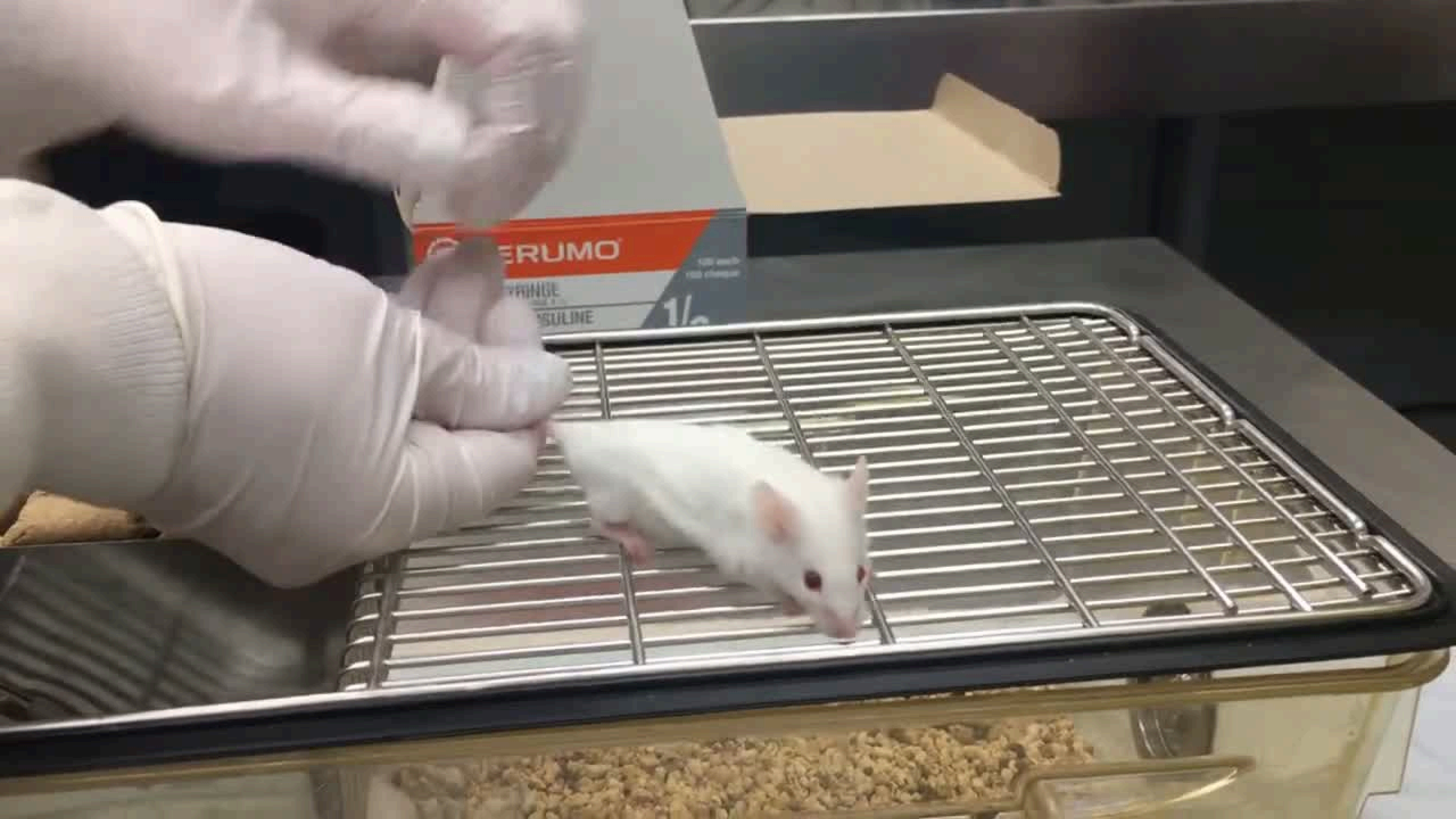
Handling of mice



Animal Injection

- **Bovine serum albumin (BSA) will be used as a soluble antigen.**
- **Incomplete Freund's adjuvant (I.F.A) will be used as an adjuvant to enhance the immune response.**
- **The proportion of the BSA-I.F.A mixture should be (1 : 1).**
- **BSA-I.F.A mixture will be injected into the mice using the intraperitoneal route. The injected dose is around 200 µl.**
- **After the first injection, you should wait at least three days before injecting the following doses.**





Canva Link

- Canva Presentation:
https://www.canva.com/design/DAGgawCuvv4/xOUqKz9BXmOfKXQwSR8pRQ/edit?utm_content=DAGgawCuvv4&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton



**BIG
THANKS**

